

[PLACEHOLDER TITLE – draft once
Results/Discussion are finalized; avoid
overlapping language with Dai et al. 2025’s
”four-hour thunderstorm nowcasting” framing, see
FINDINGS.md F3b]

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Abstract

[PLACEHOLDER – not yet drafted, see manuscript/sections/abstract.md for the planned structure. Draft last, after Results/Discussion are finalized, per standard practice – the abstract summarizes the paper, so writing it first risks having to rewrite it anyway.]

Keywords: lightning nowcasting, diffusion model, probabilistic forecasting, satellite remote sensing, Central Africa, deep learning

1 Introduction

[PLACEHOLDER – not yet drafted. Needs citation verification first (lightning hazard statistics, Central Africa lightning climatology) – see manuscript/citations_ledger.md’s ”not yet searched” list. manuscript/sections/introduction.md has the planned outline/flow. Requires an explicit differentiation paragraph against Dai et al. 2025 (FINDINGS.md F3b) – that paper has only been seen via abstract/search snippets so far, needs a full read before this paragraph is drafted.]

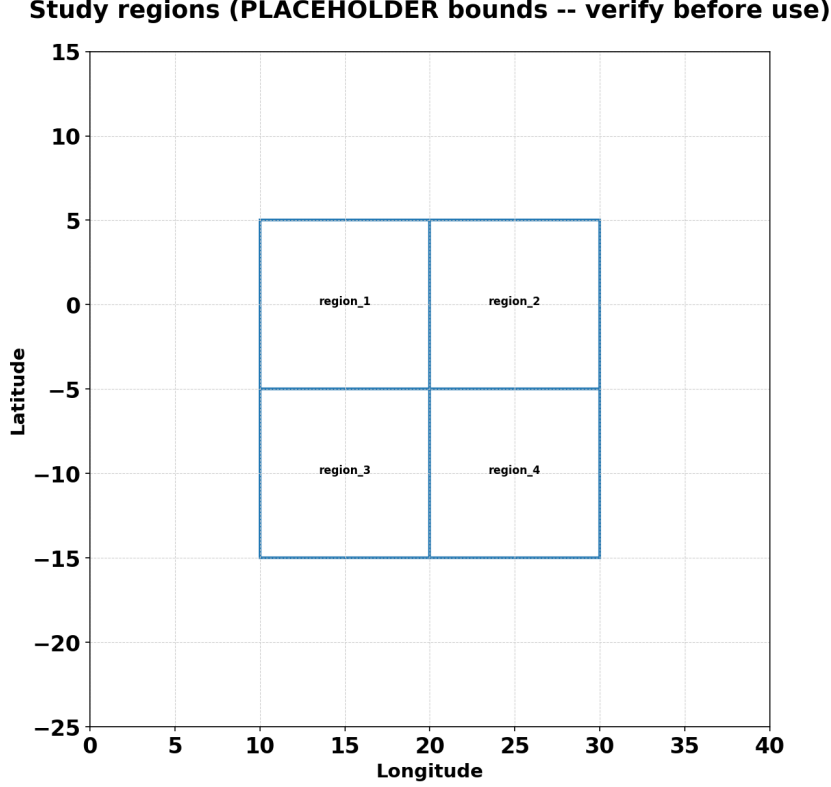


Fig. 1 Study overview. (a) Study region: four test areas within Central Africa. (b) Model schematic: 6-hour observation context processed by the EDM diffusion model, sampled as an ensemble of physically plausible forecast realizations.

2 Results

We developed a probabilistic lightning nowcasting system based on the elucidated diffusion model (EDM) framework [1], trained on infrared brightness temperature and lightning-imager observations from the MTG-FCI geostationary instrument over four regions of Central Africa. The model takes a 6-hour observation history (36 frames at 10-minute resolution) as context and generates a probabilistic forecast of lightning occurrence over the following hour (six 10-minute steps), sampled as an ensemble of 50 physically plausible realizations per forecast (Fig. 1).

The model substantially outperforms physical extrapolation baselines, with an advantage that grows with lead time

We compared the model’s pixel-wise precision-recall AUC (PR-AUC) against three physically-motivated baselines: persistence, and two variants of optical-flow extrapolation (pySTEPS-style semi-Lagrangian advection [2]) driven respectively by the lightning-imager and infrared channels. Across the six 10-minute lead-time steps tested

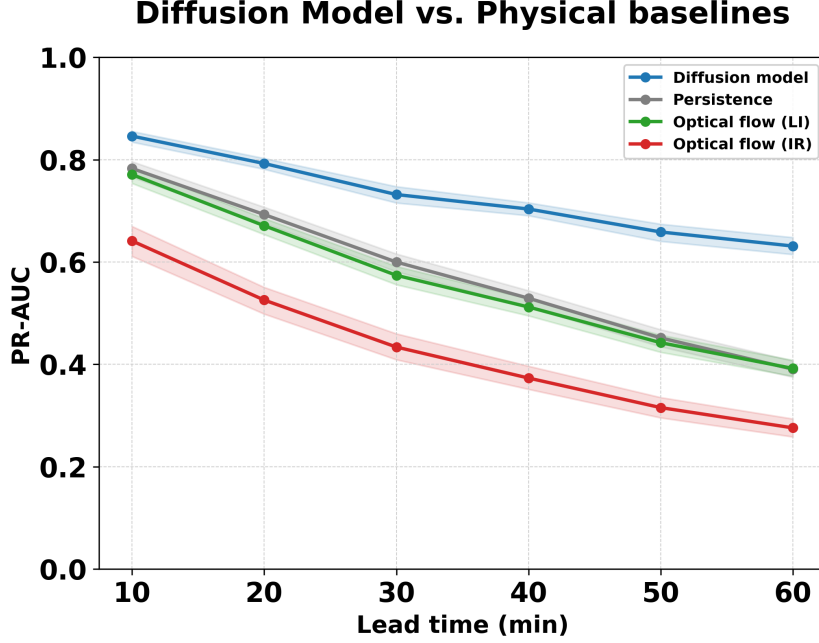


Fig. 2 Headline performance. PR-AUC vs. lead time for the model and the three physical baselines (persistence, LI-derived and IR-derived optical flow), with 95% sequence-level paired bootstrap confidence bands.

(+10 to +60 minutes), the model’s PR-AUC ranged from 0.846 to 0.631, compared with 0.783 to 0.391 for persistence — the strongest of the three physical baselines at every lead time except +60 minutes, where the LI-derived optical-flow baseline narrowly exceeded it (0.392 versus 0.391). The model’s margin over the best available physical baseline grew from 8.1% at +10 minutes to 61.0% at +60 minutes (Fig. 2).

Using sequence-level paired bootstrap confidence intervals (1,000 resamples, resampling whole test sequences rather than individual pixels to respect within-scene spatial correlation), every one of these comparisons was statistically significant (95% CI excluding zero) at every lead time tested, with the margin growing monotonically for all three baselines (persistence: +0.063 to +0.241; LI-derived optical flow: +0.075 to +0.239; IR-derived optical flow: +0.205 to +0.355; Fig. 2).

Lightning displacement at convective scale is not captured by any coherent motion field

If residual positional error were explained by a single coherent motion field — the entire forecast scene displaced by one translation vector, as assumed implicitly by any advection-based extrapolation method — then scoring each forecast against the best-possible single global shift of the observed field should recover most of the skill unlocked by tolerating position error generally. We tested this directly by comparing exact pixelwise PR-AUC against (i) the best achievable score under an unconstrained

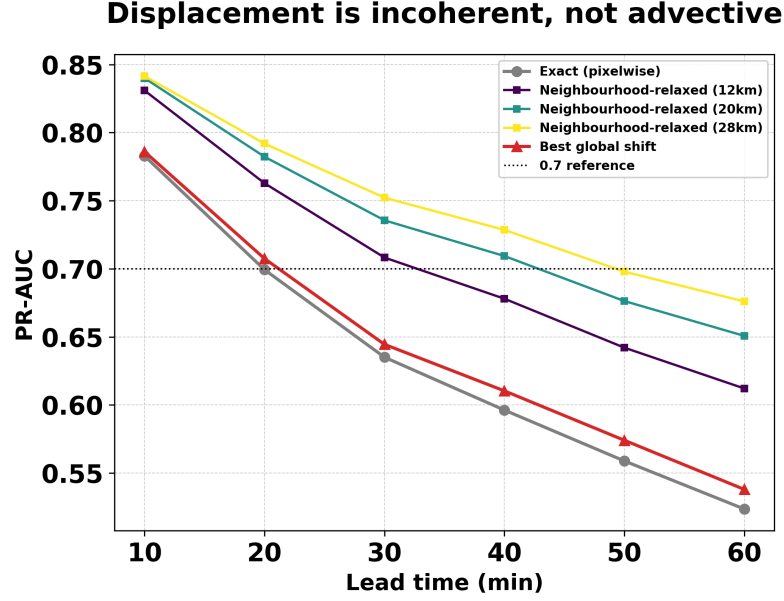


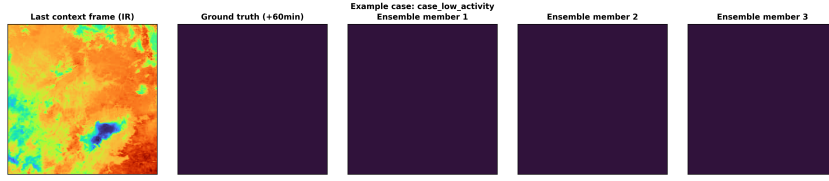
Fig. 3 Displacement is incoherent, not advective. PR-AUC vs. lead time under increasing spatial tolerance: exact pixelwise scoring, neighbourhood-relaxed scoring at several radii, and the best-case global-shift upper bound. Convergence of the relaxed and best-shift curves toward, but not beyond, a common ceiling indicates positional error is not explained by a single coherent motion field.

per-scene global shift and (ii) scores under local neighbourhood tolerance (spatial pooling at 12, 20, and 28 km radii, with no shift constraint), across sequences from the test set.

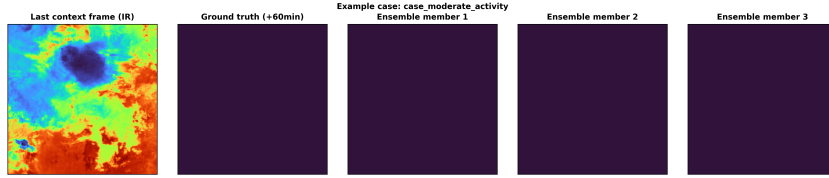
The best global shift recovered only a small fraction of the skill available under local spatial tolerance, and this fraction did not increase with lead time. At +10 minutes, allowing the single best global shift improved PR-AUC from 0.783 to 0.786 — 7% of the gain available from 12 km neighbourhood tolerance alone (0.831). At +60 minutes, the corresponding improvement was 0.524 to 0.538, or 16% of the 12 km-tolerance gain (0.612), and a substantially smaller fraction of the gain available at 28 km tolerance (0.676; Fig. 3). If the model’s positional errors were predominantly a coherent, correctable displacement, the best-shift curve should approach the neighbourhood-relaxed curves as tolerance radius increases; instead it remains close to the exact, unrelaxed score at every lead time tested. This is consistent with positional error that is spatially heterogeneous at the pixel level rather than a single global translation, and provides direct, model-independent evidence for why advection-based extrapolation baselines (persistence, optical flow) cannot close the gap to the model at these lead times (cf. Fig. 2): there is no single motion field for them to estimate.

Example forecasts illustrate calibrated ensemble diversity

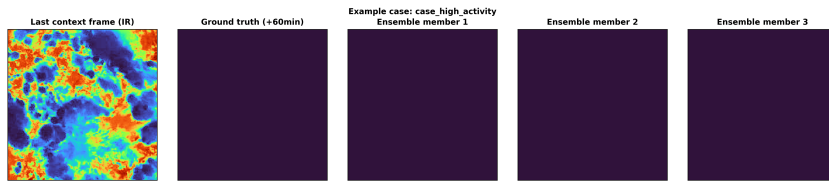
[PLACEHOLDER — checked the actual committed
fig4.example.case_{low,moderate,high}_activity.png files directly (2026-08-21): all



(a) Low-activity case



(b) Moderate-activity case



(c) High-activity case

Fig. 4 Example forecasts. Three representative test sequences spanning the observed range of lightning activity, each showing the final observed context frame, ground truth at +60 minutes, and individual ensemble member samples. [PLACEHOLDER: case selection here is automatic (ranked by ground-truth activity level), not necessarily the most illustrative uncertain/multimodal or failure case – see `select_example_cases.py`'s `--candidate_indices` option and MANUSCRIPT.PLAN.md's Figure 4 notes for manually curating a better set once real output has been inspected.]

three currently show a real, structured context frame but a completely blank ground-truth panel and blank ensemble members. This matches the exact symptom of a bug already found and fixed in `select_example_cases.py` this session (a double application of the physical-space transform on the li channel – see git history) – these three files were generated before that fix and need to be regenerated, not just described from what's currently committed. Re-run: `python select_example_cases.py --config configs/evaluate.yaml --checkpoint <path> --output_dir manuscript/figures --n_scan 60 --n_members 50`, then this subsection can be drafted from the real, corrected output. Automatic case selection (ranked by ground-truth activity) may also not be the most illustrative set – consider `--candidate_indices` once corrected output can be inspected, per MANUSCRIPT.PLAN.md's Figure 4 notes.]

A directly-optimized deterministic classifier retains a narrow, lead-time-independent pointwise advantage, explained by information access rather than model class

To assess how the model compares against directly-optimized machine learning baselines — not just physical extrapolation — we trained a convolutional neural network (CNN) baseline, architecturally and in training recipe matched to Metzl et al. [3], and a LightGBM [4] baseline using hand-engineered features, in the methodological spirit of Song et al. [5]. Both baselines’ respective training objectives (binary cross-entropy for the CNN, logistic loss for LightGBM) directly match the pointwise verification metric used here, unlike the diffusion model’s own training objective.

The model significantly outperformed LightGBM at every lead time, with a margin that grew from +0.018 at +10 minutes to +0.185 at +60 minutes — the same qualitative shape as the three physical baselines. Against the CNN baseline, however, the model showed a small but statistically significant pointwise disadvantage at every lead time (-0.021 to -0.024), and unlike every other comparison, this margin remained essentially constant across the full 50-minute range tested rather than growing (Fig. 5b; see Discussion for interpretation of this lead-time-independent shape).

This pointwise disadvantage against the CNN baseline was concentrated at loose probability thresholds and did not persist under spatial tolerance: at a strict, majority-consensus probability threshold ($p > 0.5$), the model’s Fractions Skill Score matched or exceeded the CNN baseline’s at spatial scales beyond approximately 130 km (Fig. 5a). We further found that this pointwise gap was substantially, though not entirely, attributable to the ensemble size used to construct the model’s per-pixel probability estimate: increasing the ensemble from 10 to 50 members closed roughly 60% of the original PR-AUC gap against the CNN baseline, with a clearly diminishing-returns pattern (49% of the gap closed increasing from 10 to 30 members, a further 21% of the remainder closed from 30 to 50; Fig. 5b).

The model provides calibrated, genuine forecast uncertainty

[PLACEHOLDER — draft once the reliability diagram and CRPS/spread-skill numbers from Figure 6 are finalized against real (not synthetic) data. Key point to make here: this is the capability class the CNN/LightGBM baselines structurally cannot be evaluated on, since they produce a single deterministic output rather than a sample-based ensemble.]

3 Discussion

[PLACEHOLDER — not yet drafted in this file. A substantial existing draft covering the baseline-verification portion (the F4 story) exists at manuscript/discussion_cnn_baseline_comparison.md, written before these LaTeX/citation rules were established — needs merging in here and its citations re-verified against citations_ledger.md before use. The remaining Discussion content (interpreting the incoherent- displacement finding, limitations, future work, broader significance) is outlined but not drafted in manuscript/sections/discussion.md.]

*Figure pending — generate via manuscript/generate_figures.ipynb
(fig5a_fss_vs_scale.png not yet generated)*

(a) FSS vs. spatial scale, diffusion model vs. CNN baseline

*Figure pending
(fig5b_n_members_sensitivity.png not yet
generated)*

(b) Ensemble-size sensitivity

Fig. 5 Why baseline comparison needs care. (a) Fractions Skill Score vs. spatial neighbourhood scale, model vs. CNN baseline, at three probability thresholds – the model’s disadvantage at loose thresholds does not persist under spatial tolerance, and reverses at strict thresholds beyond ~ 130 km. (b) PR-AUC vs. lead time at ensemble sizes of 10, 30, and 50 members, showing the diminishing-returns pattern underlying the ensemble-size choice used throughout this study.

4 Methods

Study region and data

[PLACEHOLDER: MTG-FCI instrument citation not yet verified — see citations_ledger.md "not yet searched" list. Do not finalize this paragraph until that citation is resolved.] The study region comprises four test regions within Central Africa (see Fig. 1a for exact boundaries). Input data are drawn from the MTG-FCI

*Figure pending — generate via manuscript/generate_figures.ipynb
(fig6_uncertainty_quantification.png not yet generated)*

Fig. 6 Genuine forecast uncertainty. (a) Reliability diagram, one line per lead time. (b) Continuous Ranked Probability Score and spread-skill ratio vs. lead time.

geostationary satellite instrument at 4 km spatial resolution and 10-minute temporal resolution, using two channels: an infrared window channel (ir) and a lightning-imager-derived channel (li). The model takes a 6-hour context window ($T_{in} = 36$ frames) and forecasts a 1-hour horizon ($T_{out} = 6$ frames at 10-minute steps).

Preprocessing

Both channels are normalized (z-scored) prior to input. The li channel additionally undergoes a cube-root transform prior to normalization, chosen because the transform preserves exact zero values — physically meaningful for a channel that is either exactly zero (no lightning activity) or continuously nonzero — which matters for the thresholding convention used consistently throughout evaluation (ground-truth binarization is always performed in physical space, after inverting both the z-score and the cube-root transform, using a strict greater-than-zero threshold for the raw physical field and a small fixed threshold, $5/255$, for probability-style outputs derived from learned models).

Model architecture

The model follows the elucidated diffusion model (EDM) framework of Karras et al. [1], implemented as a UNet built from residual blocks, conditioned on the observation context and on lead time via a shared embedding pathway. [PLACEHOLDER: confirm exact base_channels/channel_mults/num_res_blocks/attn_resolutions values against the actual training configuration before finalizing — these should be pulled directly from configs/default.yaml or the final checkpoint’s saved args, not from memory.]

Training

The model was trained with distributed data parallelism across two GPUs, using exponential moving average (EMA) weights for inference, automatic mixed precision, and a cosine annealing learning rate schedule, for 790 epochs on a regenerated, quality-controlled version of the training dataset.

Ensemble size

Forecasts are generated as an ensemble of sampled realizations from the diffusion process. We used 50 ensemble members for all final reported results, a setting chosen via a controlled sensitivity sweep (10, 30, and 50 members) that showed a clear diminishing-returns pattern: tripling the ensemble from 10 to 30 members closed roughly half of an initially large verification-metric gap against a directly-optimized deterministic baseline, while a further increase to 50 members closed only a further fifth of the remainder. We attribute the closed portion of this gap to a finite-sample bias in the empirical, ensemble-derived probability estimate at loose classification thresholds, confirmed directly by comparing each model’s predicted-positive area fraction against the true event area fraction at matched thresholds (see Discussion).

Baselines

Persistence. The most recent observed lightning field is used unchanged as the forecast at every lead time.

Optical flow. Two pySTEPS-style [2] semi-Lagrangian extrapolation baselines, driven respectively by lightning-imager-derived and infrared-derived motion fields.

Convolutional neural network (CNN). A deterministic UNet-style classifier trained directly on next-frame lightning occurrence via per-pixel binary cross-entropy, following the general architecture and training-recipe choices of Metzl et al. [3] (weight decay and learning-rate schedule matched directly to their reported values). Two departures from their exact protocol were made deliberately: the CNN baseline shares the diffusion model’s own 6-hour context window and single-model lead-time amortization (rather than Metzl et al.’s ~ 30 -minute context and separately-trained per-lead-time models), to hold input information and model structure constant across our own internal comparisons — the more rigorous choice for isolating model architecture as the controlled variable in this study specifically, even though it diverges from their literal setup.

LightGBM. A gradient-boosted-tree classifier [4] trained on 18 hand-engineered features derived from the same ir/li channels (temporal window statistics, trend/finite-difference features, local spatial neighbourhood aggregation, and event-recency features), in the methodological spirit of Song et al. [5] — not a literal reproduction of their protocol, which uses a different input resolution and feature set (including aerosol and reanalysis inputs not available to this study). Class imbalance was addressed via LightGBM’s native class-weighting mechanism rather than the manual resampling used for the CNN baseline, reflecting standard practice for each respective model class rather than a forced consistency between them.

Evaluation metrics

Skill was assessed via pixel-wise precision-recall AUC (PR-AUC), critical success index / probability of detection / false alarm ratio at multiple probability thresholds, the Fractions Skill Score (FSS) [6] at multiple spatial neighbourhood scales (12–260 km), and the Continuous Ranked Probability Score (CRPS) with its reliability/resolution/uncertainty decomposition [7], together with reliability diagrams and spread-skill relationships to assess ensemble calibration directly.

Statistical testing

All comparisons between the model and each baseline used sequence-level (not pixel-level) paired bootstrap confidence intervals, resampling whole test sequences with replacement (1,000 resamples) to respect within-scene spatial correlation in the data. [PLACEHOLDER: decide whether the bootstrap methodology needs its own citation beyond standard practice.]

Data availability

[PLACEHOLDER]

Code availability

[PLACEHOLDER — link to the public repository once finalized.]

Acknowledgements

[PLACEHOLDER]

References

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